

Parameter Settings

This document describes the parameter input within the Shiny app. It is intended to clarify the two different input procedures of log2 fold changes (log2FC) and (adj.) p-values.

1. General information

These settings have an impact on the number of compounds that are filtered. Depending on your data set, different combinations are justified. KEGG and MetScape tools are not sensitive to this input as they require the entire list of compounds that were detected independent of statistical significance.

The input column can either be P-values or Adj. P-values. Both values can be exported by Compound Discoverer.

Log2FC

- Represents the direction and magnitude of change between two conditions
- Log2FC of 1 means that a measured value is twice as high in one condition compared to the other

Used to:

- Determine up- or down-regulation
- Apply fold-change cutoff

P-value

- Represents the statistical significance of the observed change
- A p-value of 0.05 means there is a 5% chance that the observed result is due to random variation rather than a real effect
- Lower p-values indicate stronger statistical support

Used to:

- Filter compounds before analysis
- Weight evidence in enrichment and pathway analyses
- The app assumes (adj.) p-values are already computed and does not perform statistical testing itself

Adj. P-value

- Compound Discoverer applies Benjamini–Hochberg correction
- An adj. p-value accounts for testing many metabolites at once. It shows the chance of a result being a false positive after correction for multiple comparisons, making the findings more reliable
- Correcting for multiple comparisons means adjusting the statistical threshold because you're testing many metabolites at the same time, so that the chance of false positives doesn't increase just because of the large number of tests
- Adj. p-value cutoff: only consider metabolites with adjusted p-value ≤ 0.05 , meaning a maximum 5% chance of false positives after correcting for multiple tests

In other omics studies, a p-value cut-off is 0.05 or 0.01 and an absolute log2FC of 1 or sometimes even higher is applied. In metabolomics, this is not as straight forward, because a compound with a log2 FC of 0.75 with a low p-value can be a strong candidate, the same way as a compound with a high log2FC (≥ 2 or ≥ 3) that is not significant can give valuable information. Especially, if a compound class is affected. Therefore, 2 input options are available.

2. Settings

Define the filtering based on the experiment. Cell culture experiments might need stricter boundaries compared to clinical or mouse data.

2.1 Input option: Log2FC only

The input option filters compounds based on log2FC without considering (adj.) p-values of the corresponding log2FC.

Log2FC:

- Default is 1
- Only enter positive values – it will be symmetrically applied
- Increase to at least 5 if only significant compound IDs should be listed
- Enter 0 if all compounds should be taken into consideration

2.2 Input option: Log2FC and (Adj.) P-value

The input option filters compounds based on the combination of log2FC and (adj.) p-values.

Log2FC:

- Default is 0.75
- Increase this number to at least 1 if the effect of the comparison should be at least 2-fold

(Adj.) P-value:

- Default is 0.05
- Further decrease this number to decrease false positives, if necessary.